

Cost Allocation in Multi-Tenant AI Infrastructure

Solving the critical question: Who pays for what when multiple teams share expensive AI resources?



Why Cost Allocation Matters

When teams share infrastructure, they also share expenses. Without proper allocation, several risks emerge:

Subsidization

One team's overspending gets subsidized by others

Blind Spots

No visibility into actual infrastructure consumption

Inefficiency

Difficult budget planning leads to wasted resources



Proper allocation = fair use +
transparency

Cost Drivers in AI Infrastructure

1

Compute

GPU/TPU hours, CPU utilization, inference vs. training workloads

Example: A single A100 GPU can cost \$2-3/hour on major cloud platforms

2

Storage

Datasets, model checkpoints, feature stores, versioned artifacts

LLM training can require petabytes of storage for datasets and checkpoints

3

Networking

Data transfer between nodes/cloud regions, egress charges

Inter-region transfers can cost 10-100x more than intra-region

4

Services

Orchestration (K8s), monitoring, logging, security scanning

Often overlooked but can account for 15-20% of total infrastructure costs



Cost Allocation Models

Showback

Report usage per team without actual billing

- Creates awareness
- No financial impact
- Good starting point

Chargeback

Directly bill teams for their measured usage

- Strong accountability
- Resource optimization
- Requires accurate tracking

Shared Pool

Equal split regardless of actual consumption

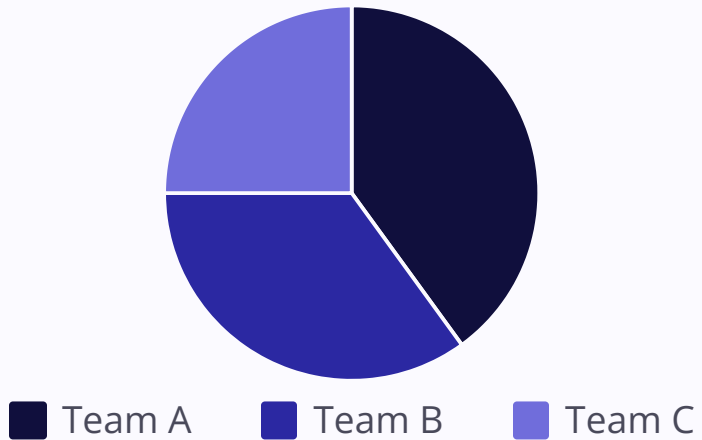
- Simple to implement
- Less fair to light users
- Potential for abuse

Hybrid

Base pool funding + usage-based overages

- Balances predictability and fairness
- More complex to administer
- Good for mature organizations

Showback in Action



Monthly GPU Usage Report

Benefits of Showback

- Creates transparency without financial pressure
- Teams see their consumption relative to others
- Identifies potential optimization opportunities
- Builds data for future chargeback models
- Drives social accountability through visibility

"You can't manage what you don't measure"

Chargeback Example

How It Works

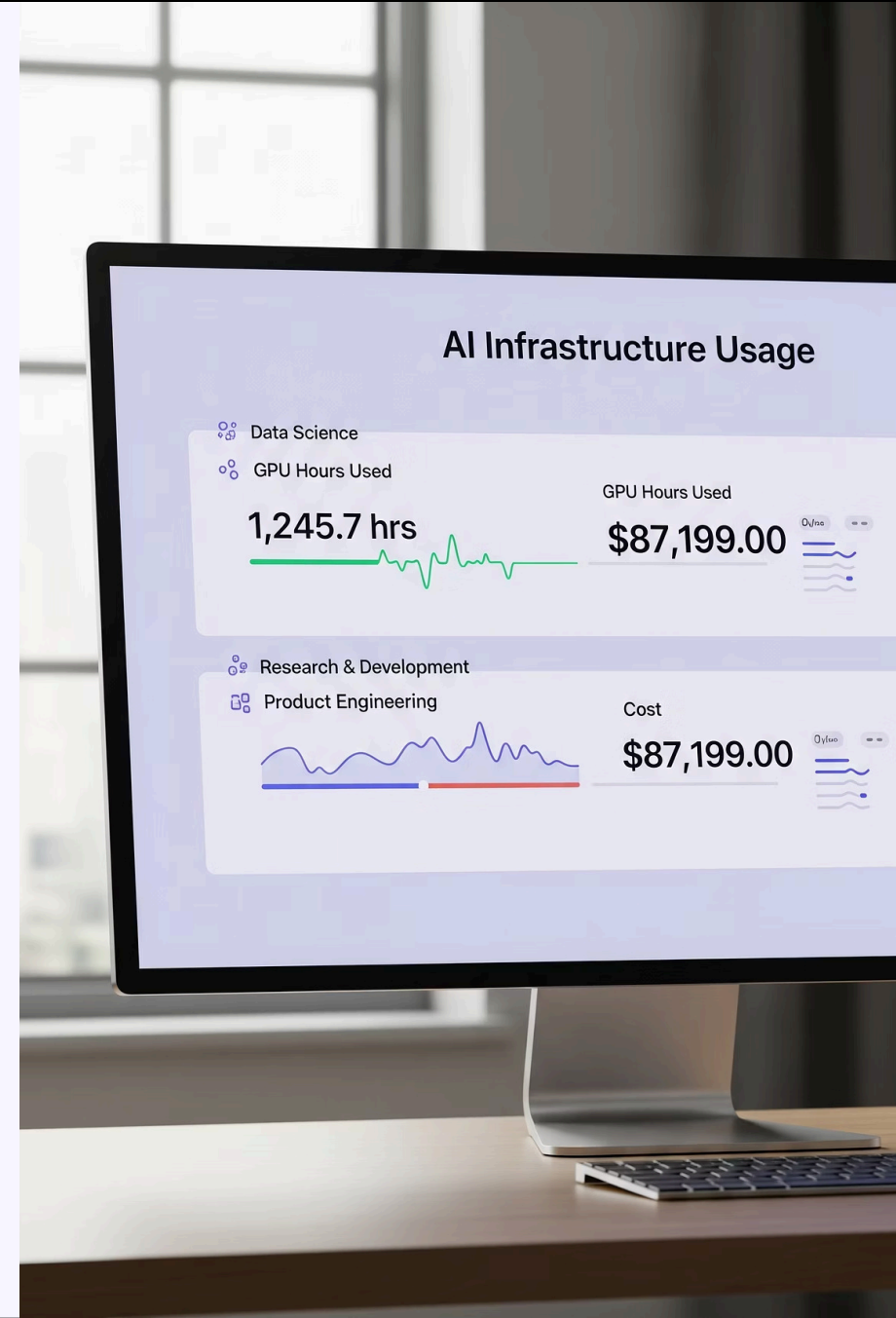
1. Tools measure GPU/CPU/Storage utilization hourly
2. Usage metrics are translated to costs (\$)
3. Teams are charged proportionally to their consumption
4. Monthly billing reports detail all resource usage

Chargeback creates a direct financial incentive for optimization

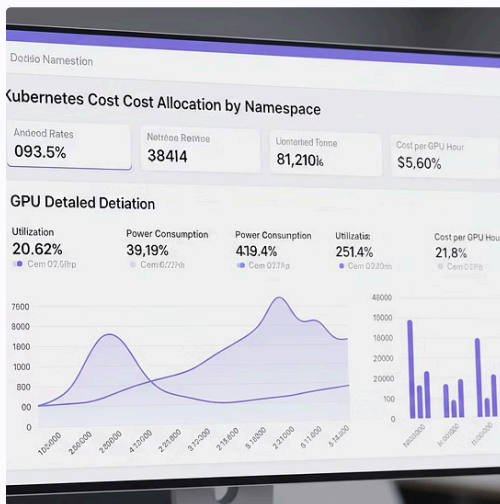
Results

Teams quickly optimize their workloads when directly billed:

- Automatic job termination when accuracy plateaus
- Right-sizing GPU allocations for workloads
- Archiving unnecessary data
- Improved model architecture efficiency



Tools for Cost Tracking



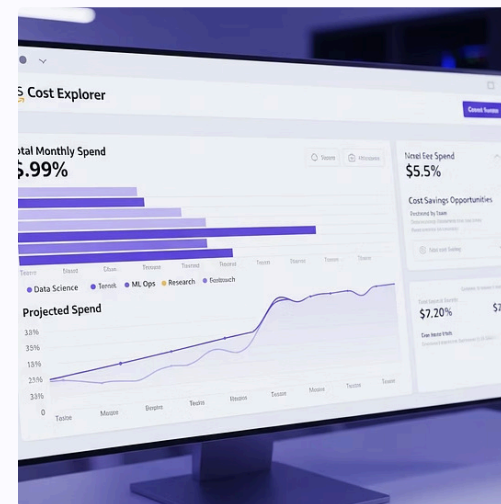
Kubecost

K8s-native cost monitoring per namespace, node, and label. Excellent for GPU tracking with custom pricing models.



Prometheus + Grafana

Open-source metrics and visualization for custom dashboards. Requires more setup but highly flexible.



Cloud Billing Tools

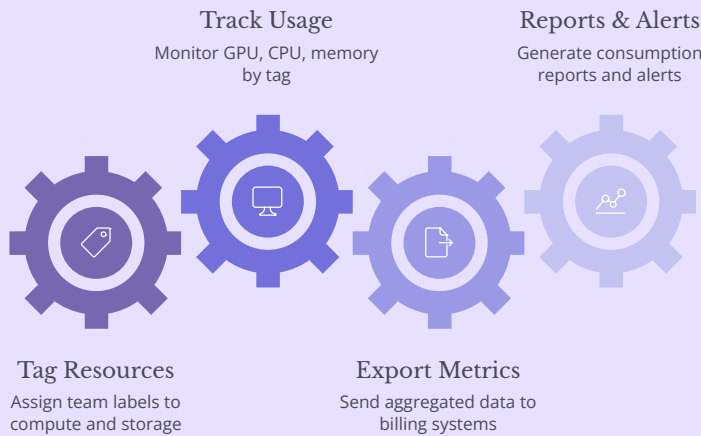
AWS Cost Explorer, GCP Billing, Azure Cost Management. Native tools for cloud-level cost visibility.



Datadog

Comprehensive monitoring with cost allocation features. Excellent for hybrid and multi-cloud environments.

Cost Allocation in Practice



Implementation Steps

1. Assign `namespace:team-a` labels to all resources
2. Configure monitoring to track usage metrics by label
3. Export aggregated data to billing or BI systems
4. Generate automated reports and consumption alerts

Required Data Points

- GPU hours by team/project/user
- Storage consumption (GB/TB)
- Memory and CPU utilization
- Job run times and completion status
- Resource idle time

Challenges in Cost Allocation



Shared Service Overhead

How do you fairly split costs for monitoring, logging, and orchestration infrastructure?



Bursty Workloads

ML training creates unpredictable cost spikes that can disrupt budgets and planning



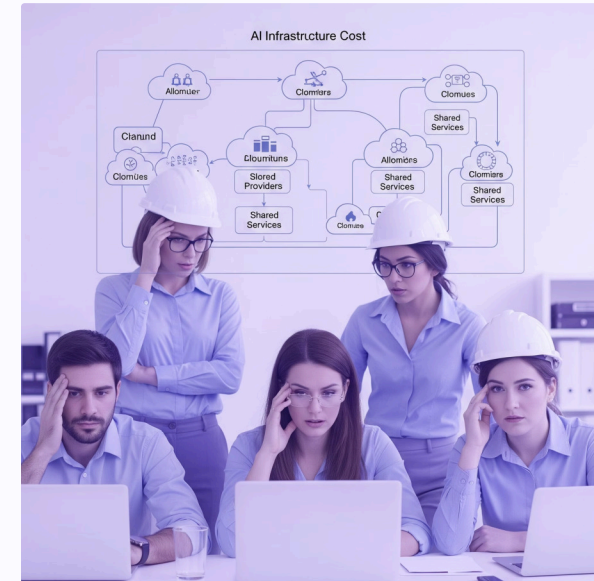
Idle Reservation

Reserved but unused GPUs waste budget—who should pay for idle capacity?



Multi-cloud Complexity

Different pricing models and metrics across cloud providers make unified allocation difficult



"The hardest part isn't tracking the costs—it's deciding what's fair."

— Senior Platform Engineer

Best Practices



Pro Tip: Create a regular forum where teams can discuss cost allocation issues and propose improvements to the model.

Real-World Case Studies

Google Vertex AI

Implements per-project billing with configurable quotas and automatic shutdown of idle instances after 1 hour.

Result: 40% cost reduction compared to always-on resources

AWS SageMaker

Uses per-user resource metering with automatic cost allocation tags. Teams define spending limits and receive alerts.

Result: Visibility into which models and datasets drive the highest costs

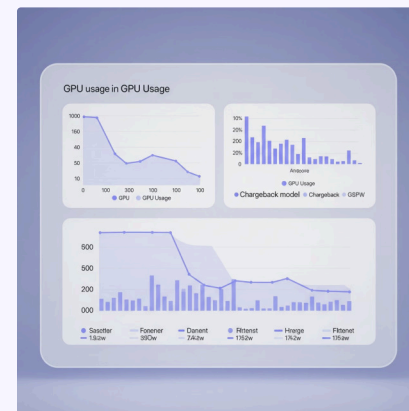
Enterprise AI Lab

Implemented Kubecost to divide GPU cluster bills across 5 departments based on actual usage rather than headcount.

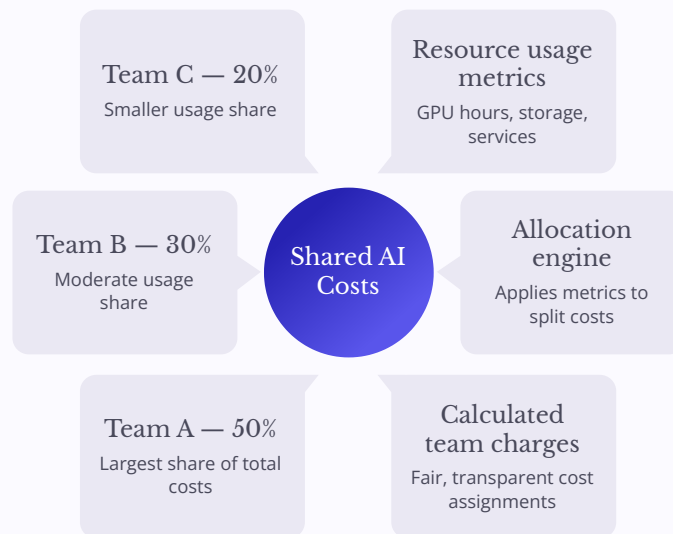
Result: 30% reduction in infrastructure costs through elimination of idle resources

"When we started charging teams directly for GPU usage, we saw a 22% reduction in total consumption within the first quarter."

— ML Infrastructure Lead at Fortune 100 Company



Cost Allocation Visualization



"Transparency is the foundation of fairness."

A well-designed allocation system creates visibility into who uses what resources, drives better decision-making around AI investments, and helps teams understand the true cost of their ML workflows.

Summary: Balancing the AI Resource Equation

Key Takeaways

- Cost allocation ensures **fairness & accountability** in shared infrastructure
- Choose the right model for your organization's maturity: showback, chargeback, or hybrid
- Implement with the right tools: Kubecost, cloud billing APIs, custom dashboards
- Balance **efficiency + transparency + fairness** in your allocation approach

Next Steps

Assess Current State

Catalog all AI resources and current cost visibility

Select Model & Tools

Choose allocation approach and implementation tools

Pilot & Refine

Start with one team, gather feedback, iterate

Questions? Contact: infrastructure-team@example.com